Radiology AI | Backend API and AI workflow integration

Frontend specification package - October 2026 | Planning baseline, not a repository audit

Contract and integration design

API boundaries - Base URL configured per environment; typed clients; request correlation IDs; standardized problem responses; explicit handling of 401/403/404/409/422/429/500; no retries for non-idempotent mutations without idempotency keys.

Example contracts - GET /api/me; GET /api/patients?search=; GET /api/worklist?status=; GET /api/reports/{id}; POST /api/reports/{id}/draft; POST /api/reports/{id}/submit-review; POST /api/reports/{id}/sign; POST /api/ai/jobs; GET /api/ai/jobs/{id}. These are proposed endpoints, not verified backend routes.

State transitions - Draft -> In review -> Signed -> Amended (with explicit version history). Frontend can display allowed actions supplied by backend; backend enforces state transitions and role authorization.

AI interactions - Use asynchronous AI jobs with job IDs, progress polling or server-sent events if supported, cancellation and errors. Clearly label unverified AI output, preserve source references and do not auto-populate signed findings.

Privacy & auditing - Mask identifiers in analytics, prevent cache persistence of patient data, restrict download/print actions, log clinical actions server-side; do not rely on frontend audit events alone.

Mocks - Use MSW fixtures for development, synthetic patients/studies, deterministic demo scenarios and contract tests. Never connect public demo builds to real patient datasets.

Domain-specific adapters

DICOM integration - Evaluate current OHIF and Cornerstone capabilities and licensing; integrate DICOMweb QIDO-RS/WADO-RS through authorized backend or PACS endpoints; never expose unrestricted PACS credentials to browsers.

Performance - Virtualize large worklists; lazy-load viewer; avoid fetching entire study into React state; memory and GPU limits tested on supported browsers; image rendering performance benchmarked with realistic de-identified studies.

Domain standards - FHIR ImagingStudy and DiagnosticReport mappings may be introduced through backend interoperability layer, not as assumptions about current APIs.

Backend coordination

Discovery checklist - Read backend OpenAPI, authentication flow, tenant IDs, status enums, pagination, AI job lifecycle, audit endpoints and clinical report signing permissions.

Contract versioning - Document DTO ownership, API versioning, nullable fields, pagination cursor or page strategy, optimistic concurrency and attachment upload size/security.